

$$2h + 3x + 5w = 400 \quad \dots (i)$$

$$3h + x + 2w = 300 \quad \dots (ii)$$

$$h + 2x + 4w = 280 \quad \dots (iii)$$

From (i)

$$2h = 400 - 3x - 5w$$

$$h = 200 - \frac{3}{2}x - \frac{5}{2}w$$

Putting h in (ii) & (iii)

$$3\left(200 - \frac{3}{2}x - \frac{5}{2}w\right) + x + 2w = 300$$

$$600 - \frac{9}{2}x - \frac{15}{2}w + x + 2w = 300$$

$$-\frac{7}{2}x - \frac{11}{2}w = -300$$

$$\frac{7}{2}x + \frac{11}{2}w = 300$$

$$7x + 11w = 600 \quad \dots *$$

$$\left(200 - \frac{3}{2}x - \frac{5}{2}w\right) + 2x + 4w = 280$$

$$+\frac{1}{2}x + \frac{3}{2}w = 80$$

$$x + 3w = 160 \quad \dots **$$

$$x = 160 - 3w$$

Putting x in *

$$7(160 - 3w) + 11w = 600$$

$$1120 - 21w + 11w = 600$$

$$-10w = -520$$

$$\therefore w = 52 \text{ (wish)}$$

From $7x + 11w = 600$,

$$x = \frac{600 - (11 \times 52)}{7}$$

$$\therefore x = 4 \text{ (marks)}$$

$$h = 200 - 6 - 130$$

$$\therefore h = 64 \text{ (hairer)}$$